



PRESIDENCY UNIVERSITY

**SCHOOL OF ARTIFICIAL INTELLIGENCE &
ADVANCED COMPUTING**

NVIDIA GPU USER SUMMER INTERNSHIP

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TITLE:

CREDIT CARD APPROVAL PREDICTION SYSTEM

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1. ABSTRACT

The "Credit Card Approval Prediction System" is a web-based application designed to provide instant decisions for credit card applications. The system collects 15 parameters from the user such as Income, Assets, Debt, Credit Score, and Employment Status. Initially a pre-trained ML model was used, but it was not responding to input changes. Therefore, a Rule-Based Scoring System was implemented. This approach makes the system transparent and ensures that changing key inputs like Income immediately affects the result. The application is developed using Python and Streamlit framework. The system displays both the decision APPROVED or REJECTED along with a probability percentage. This project is ideal for demonstration and shows how simple business rules can be used for decision making.

2. INTRODUCTION

In the modern banking sector, millions of credit card applications are processed every year. Manual verification is slow and prone to errors. There is a growing need for automated systems that can give quick and consistent decisions. Machine Learning models are commonly used, but they act as a "black box" and are difficult to explain. Also, poorly trained models may not respond correctly to changes in input data. To overcome this, we developed a

rule-based system. In this system, each financial parameter is assigned points based on standard banking logic. If the total score is above a threshold, the application is approved. This makes the system fast, transparent, and easy to demonstrate for academic purposes.

3. SYSTEM REQUIREMENTS

3.1 Software Requirements

- Operating System: Windows 10 / 11
- Programming Language: Python 3.9
- Framework: Streamlit
- Libraries: NumPy
- IDE: VS Code

3.2 Hardware Requirements

- Processor: Intel i3 or above
- RAM: 4GB minimum
- Storage: 100MB

5. IMPLEMENTATION

The project is divided into two modules.

Module 1: User Interface

Built using Streamlit. It takes all 15 inputs from the user through text boxes and dropdowns.

Module 2: Prediction Engine

This is the backend logic written in Python. It calculates the total score and displays the result.

6. TESTING AND RESULTS

The system was tested with different input values to verify the working.

Test Case 1: Low Financial Profile

Input: Income = 15,000, Assets = 20,000, Debt = 10,000

Output: REJECTED - 85% Probability

[Insert Screenshot 1]

Test Case 2: High Financial Profile

Input: Income = 250,000, Assets = 600,000, Debt = 0

Output: APPROVED - 90% Probability

[Insert Screenshot 2]

7. ADVANTAGES

- Instant Response: Result is displayed in less than 1 second.
- Transparency: The user knows exactly why the decision was made.
- No Training Required: Works without any dataset.
- Flexible: Rules can be easily modified by the bank.

8. CONCLUSION

The Credit Card Approval Prediction System successfully demonstrates a fast and transparent method for decision making. By using a rule-based approach instead of a faulty ML model, we achieved the main objective of making the output directly dependent on user input. This project is suitable for academic demonstration and prototype development.

9.Future Scope

- Integrate a real ML model trained on actual bank data.

- Add a database to store all applications.
- Implement user login and admin panel.
- Deploy the application on cloud for public access.
- Add feature to generate and download PDF approval/rejection letter.

10. LEARNING OUTCOME

This project provided hands-on experience in building a complete end-to-end web application using Python and Streamlit. It helped us understand the difference between machine learning based systems and rule-based systems in terms of transparency, debugging, and reliability. While ML models are powerful for large-scale real data, rule-based systems are more suitable for prototypes, academic projects, and scenarios where explainability is important. Working on this project improved our skills in UI development, backend logic implementation, problem-solving, and technical documentation. It also gave us insight into how real banking systems make quick automated decisions. Overall, this project serves as a strong foundation for building more advanced fintech applications in the future.